

The significance and evolution of biological databases in systems biology

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Over the years as systems biology has evolved as a field, the demand for organized and consolidated biological data has grown significantly. Databases have become crucial in enabling the storage, retrieval, and analysis of types of omics data at a scale. In this section, we provide an overview of how biological databases have progressed over time and emphasize their increasing importance in systems biology research. We will delve into the evolution of system biology databases, discuss the underlying principles behind designing database architecture strategies to address data integration challenges, explore the tasks related to curating and managing databases, and look ahead to what the future holds for databases.

1. Historical perspective of biological databases

The origins of systems biology databases can be traced back to the mid-20th century when researchers achieved groundbreaking discoveries in protein and nucleic acid sequences. In 1956, a significant milestone was reached with the determination of insulin sequence, which consists of 51 amino acids [1]. This marked a step toward collecting data on protein sequences. In addition, in 1965, researchers successfully decoded the sequence of yeast alanine tRNA which is a molecule composed of 77 nucleotide bases [2], further contributing to the growing dataset of sequences [3]. These early breakthroughs laid a foundation for the development of bioinformatics databases. In the 1970s, Margaret Dayhoff achieved a milestone in bioinformatics by creating one of the first databases for protein sequencing. Her accomplishment involved gathering existing data on protein sequencing and compiling it into a collection [4]. The establishment of the Protein Data Bank (PDB) in 1971 marked a development initially containing 10 protein structures identified through X-ray crystallography. These initial efforts laid the foundation for the creation of data repositories and advancements in analysis tools [5]. In the 1980s and early 1990s, notable progress was made in the field of bioinformatics. SwissProt, an organization founded in 1986 gained recognition for its management of protein sequences. In addition, the National Center for Biotechnology Information (NCBI) was established as a hub for data in 1988. During this time period, there was also the establishment of the Human Genome Initiative with the aim of mapping all genes. Furthermore, groundbreaking algorithms like BLAST and FASTA were introduced during this era which revolutionized sequence analysis techniques. In the 1990s, two important resources were established; Kyoto Encyclopedia of Genes and Genomes (KEGG) providing information on pathways and genomics, and Ensembl, which offers genome annotation and visualization tools [6]. The decoding of the code was an accomplishment and it marked a major milestone when the Human Genome Project successfully completed the initial version of the human genome in 2000. During that year two repositories were established: Gene Expression Omnibus (GEO), by NCBI to store gene expression data [7] and ArrayExpress by the European Bioinformatics Institute (EBI) which focused on genomics data [8]. Fig. 10.1 provides an overview of milestones and developments in databases throughout history.

These databases have played roles in bioinformatics by offering data and tools for global research on genes, proteins, pathways, and gene expression. Over time bioinformatics tools and databases have advanced from sequence searches and comparisons to computer programs and well-organized data repositories. Nevertheless, many foundational concepts from bioinformatics remain crucial today.

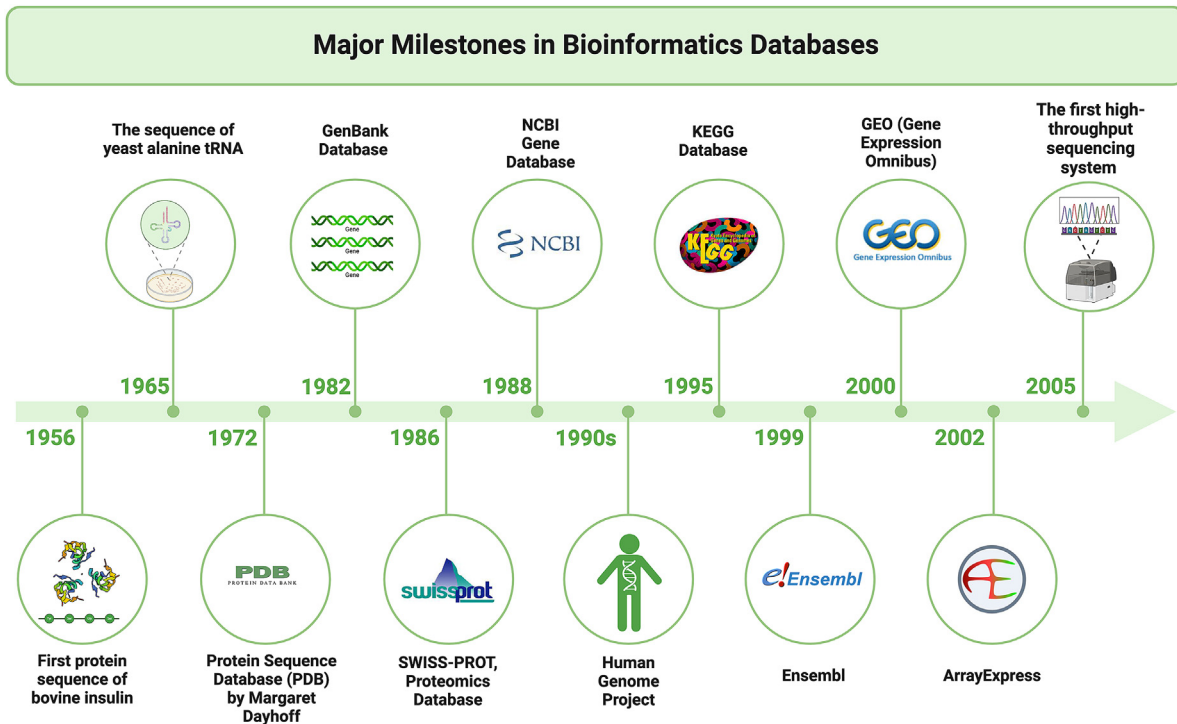


FIGURE 10.1 A historical timeline of important bioinformatics databases. Over the years there have been advancements, in the field of bioinformatics databases. These include efforts in sequencing during the mid-20th century well as the creation of valuable resources such as the Protein Data Bank (PDB) SwissProt, NCBI, KEGG, Ensembl. A significant milestone was achieved with the completion of the Human Genome Project, in 2000. This figure represents a timeline of significant bioinformatics databases. All the information within the figure is condensed under the [Section 1](#).

2. Architecture and design considerations for modern databases

In recent decades, there have been significant advancements in the architectural and design aspects of biological databases. Historically, databases primarily served as basic storage facilities for data. However, contemporary databases have evolved into intricate systems designed to optimize performance, accommodate scalability, and facilitate analytical functionalities. These changes have been driven by the emergence of high-throughput technologies, which have greatly increased the volume and complexity of data. The shift from repositories to cloud-based infrastructures represents a major transformation in the field of biological databases. With the rise of “data”, traditional database architectures have been redesigned to handle rapidly expanding datasets. The integration of cloud computing has been a game changer by providing storage and powerful computational resources that are essential for systems biology applications. In this era of rapid advancements, cloud-based databases have become essential for storing and effectively managing the amount of biological data produced by high-throughput technologies [9].

The use of cloud technology offers a wide range of storage capabilities and adaptable processing capacities, hence facilitating the growth of databases and efficient management of large-scale data inputs [10]. Major biological databases such as GenBank have undergone a shift toward cloud-based solutions [11]. Cloud databases enable users to access data as needed and conduct real-time analysis. The incorporation of artificial intelligence and machine learning techniques into biological databases has resulted in enhanced query and predictive capabilities. Currently, database search engines use natural language processing and neural networks to comprehend user queries expressed in free-text format, enabling them to provide results that are contextually relevant [12]. Algorithms provide the capability to reveal hitherto undiscovered data relationships, hence enabling novel discoveries. The use of AI, as shown by tools such as DeepDB, showcases the inherent capability to augment and optimize the many operations of databases [13]. NoSQL graph databases are capable of effectively representing intricate biological networks, such as protein interactions, gene regulation, and metabolic processes [14]. The network-based architecture facilitates the seamless retrieval of linked data via straightforward querying [15]. As another example of AI-powered databases, Neo4j has gained significant popularity as a prevalent graph database in the field of systems biology research [16].

The remarkable progress toward utilizing cloud-based solutions and incorporating AI into databases has paved the way for a range of database technologies. These advancements have not only improved the capacity of databases to handle larger amounts of data, but they have also introduced various types of databases designed to address specific needs for storing, retrieving, and analyzing data. As we delve deeper into the architecture of modern databases, it becomes clear that the choice of database type is fundamental and influenced by the nature of the data being handled, the required performance characteristics, and the desired level of flexibility in data management. On one hand, we have relational databases, which have been the heart of data storage solutions for decades. On the other hand, the explosion of big data has given rise to nonrelational databases, which offer a different approach to data organization and scalability.

2.1 Database types

Databases can be categorized into two main types: *relational*, such as MySQL, and *nonrelational*, such as MongoDB. Relational databases are designed based on the fundamental concept of tables, wherein data are systematically arranged into rows and columns. The databases employ a schema to establish the framework and interconnections among distinct data tables. The organization is known for maintaining consistent data using ACID (Atomicity, Consistency, Isolation, Durability) transactions, and provision of comprehensive query capabilities via SQL (Structured Query Language) [17]. NoSQL databases, sometimes referred to as “Not Only SQL,” are a classification of databases that provide adaptable and schema-less data storing capabilities. In contrast to relational databases, NoSQL databases do not depend on a rigid schema, enabling them to efficiently manage semistructured and unstructured data. NoSQL databases are commonly employed in situations where there is a need for flexibility in accommodating quickly changing data types or volumes. They are particularly well-suited for applications that include real-time processing of large-scale data [18]. Relational databases are designed to improve the structure and integrity of data, while nonrelational databases prioritize attributes such as flexibility, scalability, and availability; hybrid databases integrate both types.

2.2 Database models

The process of data modeling involves the creation of a conceptual representation of data structures and their interconnections inside a database. The process entails the establishment of the framework, limitations, and principles that dictate the organization and retrieval of data. Data modeling plays a crucial role in comprehending the data needs of an application, guaranteeing the integrity of data, while allowing its effective management [19]. There exist multiple database models, each tailored to unique utilization scenarios. The *hierarchical model* is a data organization approach that arranges information in a hierarchical structure like a tree, characterized by parent–child relationships. This model finds widespread application in file systems and certain early database systems. The *network model* expands upon the hierarchical model by permitting various connections between records, thus providing greater adaptability but also introducing heightened intricacy. The *entity-relationship model* (ER Model) is a graphical representation that depicts entities, which are objects or concepts, and their relationships. It serves as a fundamental tool for the design of relational databases [20]. Many database models offer distinct methodologies for structuring and administering data, and the selection of a model is contingent upon the precise demands of the application or system under development.

2.3 Schema design for biological data

A *database schema* refers to a conceptual model or architectural framework that outlines the structure and organization of data in a database system. The components encompassed within this framework consist of tables, fields, data types, restrictions, and interconnections within tables. The schema provides a systematic representation of the data model and serves as a framework for data management. Furthermore, it serves the purpose of maintaining data integrity and ensuring consistent storage and accessibility of data [21]. The construction of a well-defined schema is crucial for maintaining data accuracy and consistency in a database.

Data modeling design principles encompass a set of guidelines. The foundation of effective database management lies in a series of fundamental principles and practices. These key elements, including:

1. **Normalization** is a process that involves organizing data in a way that minimizes redundancy and enhances integrity. It entails dividing data into tables to reduce anomalies.
2. **Data integrity** ensures accuracy, consistency, and adherence to established constraints and standards.

3. The ability of a database system to meet application demands relies heavily on its **scalability**. Vertical scalability involves increasing resources within a server while horizontal scalability spreads data across servers to handle incoming requests.
4. **Flexibility and maintainability** are key aspects of database design. It is essential to create database models that can accommodate changing data requirements and objectives without disrupting the application.
5. **Performance optimization** in databases focuses on improving speed and efficiency in areas such as data retrieval, storage, and query processing.
6. Ensuring the **security** of data is extremely important, especially considering the increasing number of data breaches. To minimize the chances of access, it is crucial to incorporate security measures into the database architecture that is being built.
7. **Documentation** plays a [22] role in improving understanding, simplifying maintenance, and fostering teamwork by creating a detailed record of the data model [20].

The proficient implementation of these design principles is necessary to develop a data model that fulfills the requirements of the users, is sustainable, and achieves optimal performance.

While the principles of schema design ensure that databases can handle structured data with high integrity and performance, the dynamic nature of omics data poses new challenges. Traditional relational databases, structured around predefined schemas, often fall short when confronted with the scale and heterogeneity of biological data [23]. Emerging database technologies, such as data lakes, offer a solution by accommodating the diverse and rapidly expanding datasets typical in systems biology. The use of data lakes has increasingly become a common approach for consolidating large volumes of multiomics data, often in their native file formats. Data lakes provide schema-on-read, a procedure that applies structure during the querying phase, rather than imposing schemas beforehand. This concerns the need of effectively handling unstructured and semistructured data [24]. Cloud storage providers such as Amazon Web Services (AWS) have substantial storage capacity and scalable processing capabilities, making them well-suited for the establishment of data lakes. Query engines provide the functionality to perform direct analysis on dataset dumps that are kept inside the data lake [22]. Nevertheless, the utilization of data lakes presents significant data governance obstacles pertaining to security measures, metadata administration, and the assurance of data quality.

The area of systems biology has also been greatly influenced by NoSQL database management technologies, such as MongoDB, Neo4j, and Cassandra. These technologies have had a considerable impact due to their ability to horizontally scale and effectively handle unstructured data models. The document data type provided by MongoDB facilitates the efficient storage of complex JSON-like documents, making it very suitable for the storage of omics data [25]. The Neo4j graph database has remarkable efficacy in the administration of complex biological networks, including pathways and protein interactions. Wide-column databases, such as Cassandra, provide efficient and scalable methods for effectively handling time-series omics data, exhibiting remarkable performance in web-scale systems [23].

3. The role of databases in integrating multiomics data

The term “*omics*” encompasses a range of disciplines within the subject of biology that share the common suffix “-omics,” including genomics, transcriptomics, proteomics, and metabolomics. The field of omics endeavors to comprehensively analyze and measure biological molecules that contribute to the organization, operation, and changes within an organism. The field of genomics encompasses the comprehensive examination of an organism’s complete genome, which involves the processes of DNA sequencing and subsequent analysis. Transcriptomics is a field that focuses on the study of the transcriptome, which encompasses the entirety of RNA transcripts generated by an organism’s genome [26]. Proteomics is a discipline that involves the comprehensive analysis of the complete set of proteins expressed by an organism’s genome. Metabolomics focuses on the study of metabolites and low molecular weight intermediates that play roles in metabolic reactions. Additional omics fields include, but are not limited to, epigenomics, microbiomics, and interactomics. The integration of multiomics data involves the amalgamation of datasets from many omics levels to provide a full and holistic comprehension of biological processes at the system level [27].

The integration of data originating from several omics domains allows us to understand complex biological systems. This helps us gain insights into the aspects by analyzing various datasets. For example, when we incorporate variations of gene expression patterns into protein and metabolite levels, we can uncover interconnected associations. Biological functions arise from the interactions between components. To fully understand these connections researchers integrate omics data by combining information from different extensive datasets. By taking a system-level approach, we can discover processes and regulatory pathways that may not be easily identified by analyzing individual omics datasets.

Therefore, the integration of multiomics data is of utmost importance to achieve a comprehensive comprehension of disease causes, pharmacological effects, microbial communities, and other intricate systems [28].

The integration of multiomics data is strongly dependent on the utilization of databases to ensure efficient data management. Specialized repositories are responsible for the storage, organization, and annotation of a wide range of omics datasets. Standardized IDs serve the purpose of establishing connections between different sorts of data. Advanced querying interfaces enable the integrated examination of interconnected data. Another critical category, ontologies, facilitates the mapping of items across several domains, hence offering contextual information. Semantic integration and data mapping facilitate the establishment of meaningful linkages between heterogeneous sources. As user-friendly tools, Application Programming Interfaces (APIs) enable the automated access to computational workflows. Cloud infrastructure has the capability to accommodate large quantities of data by providing scalability [29]. The practice of master data management is implemented to ensure the maintenance of consistency across many systems. To further ensure efficient data management, the utilization of data compression techniques plays a crucial role, conserving storage capacity while preserving the integrity of the information. In tandem with this, the concept of versioning is critical, as it enables the tracking of data provenance over different stages or iterations, thus maintaining a clear historical record of data changes. In the realm of interconnected multiomics big data, the significance of thorough database architecture cannot be overstated. It is essential for facilitating findability, accessibility, and unified analysis of such data [30]. As shown in Fig. 10.2, a comprehensive data flow diagram outlines the steps involved in integrating omics data into a central database [31].

Table 10.1 also provides a summary of key omics databases and resources including the omics technologies they encompass. These databases play a role, in storing and granting access to a range of omics data supporting research in genomics, transcriptomics, proteomics, epigenomics, metabolomics, and other related fields [32,33].

The integration of multiomics methodologies has become prevalent in the domain of systems biology, with noteworthy instances encompassing:

- The National Institutes of Health (NIH) Common Fund launched the Library of Integrated Network-Based Cellular Signatures (LINCS) study, in 2011. The main objective of this initiative is to collect and analyze data to improve our understanding of biology as a whole. This is done by monitoring changes in gene expression and other biological processes that occur when different chemicals affect systems. The LINCS primarily aims to comprehend how cells respond to genetic and environmental stressors. To achieve its goals, LINCS employs an omics approach generating diverse datasets that include transcriptomic, proteomic, and cell imaging data. These datasets come from groups of cells

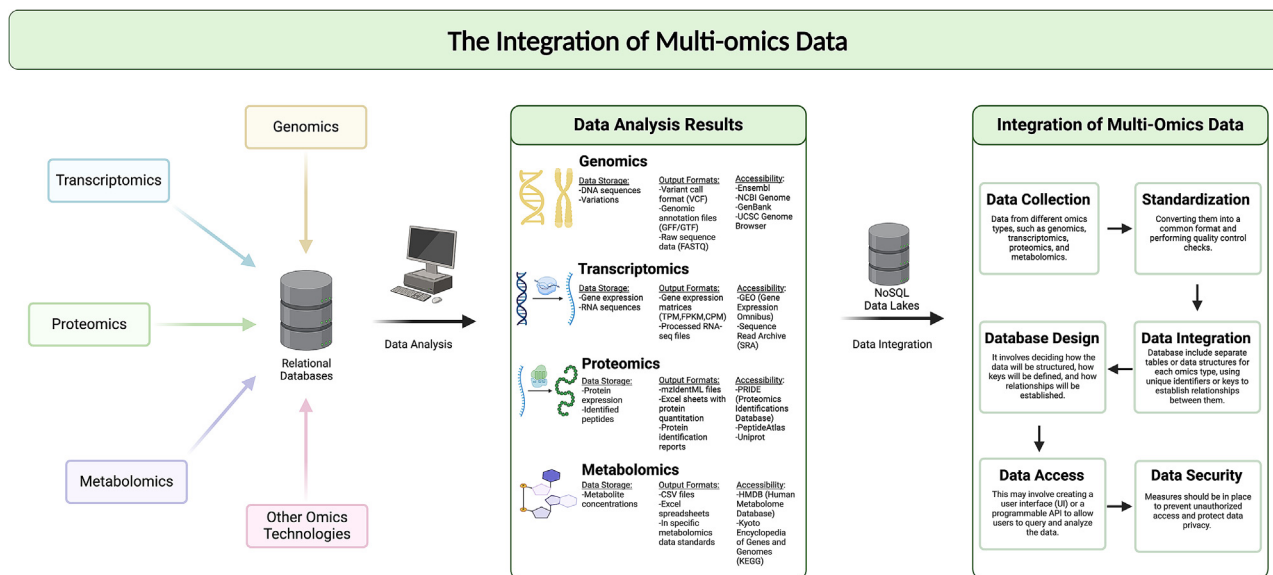


FIGURE 10.2 Data flow of the multiomics data integration. This diagram illustrates the steps involved in combining types of omics data. The data includes genomics, transcriptomics, proteomics, metabolomics, and other relevant datasets which are all merged together in a database. To achieve this integration, thorough analysis is conducted using tools. The results obtained are then effectively consolidated into databases using technologies, like data lakes and nonrelational databases. Throughout this process, it is crucial to follow principles, such as standardizing the data and implementing security measures. This figure represents a data flow of analysis and integration of multiomics data. All the information within the figure is condensed under the Section 2.3.

TABLE 10.1 List of multiomics databases.

Databases	Omics-technologies	Description
Genomic Data Commons (GDC)	Genomic Transcriptomic Proteomic Epigenomic	GDC is a database that houses a wide range of omics data for cancer research.
The Cancer Genome Atlas (TCGA)/Clinical Proteomic Tumor Analysis Consortium (CPTAC)	Genomic Transcriptomic Epigenomic Proteomic	TCGA and CPTAC are two major collaborative research initiatives in the field of cancer research.
European Bioinformatics Institute (EBI)	Genomic Metagenomic Transcriptomic	EBI offers a wide range of omics data resources, including European Nucleotide Archive (ENA), UniProt, and InterPro.
Genotype-Tissue Expression (GTEx) Project	Genomic Transcriptomic	GTEx contains genomic and gene expression data from various human tissues.
The Kyoto Encyclopedia of Genes and Genomes (KEGG)	Genomic Transcriptomic Proteomic Metabolomics	KEGG is a widely used and comprehensive bioinformatics resource that provides information on biological pathways, diseases, drugs, and genomic information.
Plant Metabolic Network (PMN)	Genomic Proteomic Metabolomics	PMN is a comprehensive online resource and database that focuses on plant metabolism.
Omics Discovery Index	Genomic Transcriptomic Proteomic Metabolomics	OmicsDI is a web-based platform and resource that integrates and indexes data from various omics disciplines.
Sequence Read Archive (SRA)	Genomic Metagenomic Transcriptomic	SRA is a repository for high-throughput sequencing data.

exposed to various treatments. By analyzing networks using these omics datasets, researchers can explore how perturbations affect genes, proteins, and biological processes. Network modeling methods are used to combine the omics datasets enabling valuable insights into cellular information flow and functionality. Due to its ability to generate profiles across different omics layers, LINCS provides a valuable resource to researchers in the field of systems biology. By combining these profiles, researchers can acquire a more integrated comprehension of biological reactions and pathways related to diseases [34].

- The National Cancer Institute (NCI) and the National Human Genome Research Institute (NHGRI) started The Cancer Genome Atlas (TCGA) project in 2005. The main goal of this project was to comprehensively list the basic genetic changes that occur in 33 different types of cancer. Researchers from TCGA investigated tumor samples from more than 11,000 patients. Several genome-wide tools were used in this study, including whole exome and whole genome sequencing, RNA sequencing, microarray-based gene profiling, methylation profiling, and protein analysis. Multiomics approach made it easier to look at and connect all the biomolecular processes that are defective in cancer cells. Multiomics study of TCGA data has made it easier to make several important findings. Finally, in 2018, the TCGA was replaced by what is now called the Genomics Data Commons (GDC). This change was made to help the project reach its goal of making cancer genomics data easier to share and study in a way that is open to everyone. In addition to the basic TCGA datasets, the GDC also hosts imaging, pathological, and clinical data now. As a result, we can get a better picture of how cancer starts and progresses at the molecular level with this wider range of data sources [35].
- The Human Microbiome Project (HMP) aims to explore the world of the human microbiome. This includes studying bacteria found in parts of our body, such as the digestive system, skin, and mouth. The HMP uses research approaches like genomics, metagenomics, transcriptomics, proteomics, and metabolomics to understand how these microbial communities interact with health. By studying the makeup, functions, and metabolism of the microbiome, the HMP aims to

uncover its impact on physiology and diseases. This initiative plays a role in advancing our understanding of how the microbiome affects health and has significant implications for personalized treatment options, nutrition recommendations, and disease prevention measures [36].

- While countless numbers of multiomic studies can be found in the literature, some of these are as follows: The study conducted by Zhanget et al. provides information about functional potential of the human gut microbiome by integrating data from metagenomics, metatranscriptomics, metaproteomics, and metabolomics [37]. The study conducted by Pinuet et al. uses metabolomics-informed genome-scale models employ constraint-based modeling techniques to integrate genetics, gene expression, and metabolomics data, to understand the links between genotype and phenotype [38]. The study conducted by François et al. integrates proteomics and metabolomics data to provide information for discovering novel diagnostic biomarkers that can be utilized for the early identification of Alzheimer's disease [39]. The study conducted by Lau et al. integrates *Arabidopsis* molecular interaction databases to identify of pivotal gene regulation and signaling networks in the field of plant biology [40]. The study conducted by Yan et al. develops data-driven network models for the prediction of prospective therapeutic targets and suggestive biomarkers in various types of cancer [41]. The study conducted by Zampieri and Sauer explores intricate interplay between genotype, gene expression, and metabolic characteristics in microorganisms by the utilization of multiomics analysis [42]. The study reported by Sen and Oresic integrates multiomic data for the reconstruction of genome-scale metabolic models [43].

These findings provide evidence for the significance of integrating multiomics data to understand the mechanisms and interactions inside intricate biological systems and demonstrate the potential benefits of integrating data from diverse public databases. To do so, advanced bioinformatics methodologies are utilized, resulting in valuable knowledge and significant insights that cannot be obtained from individual datasets alone.

4. Challenges faced in database management and potential solutions

The management and utilization of biological databases provide a multitude of significant challenges. Biological databases encounter such challenges across five primary domains.

4.1 Data quality

The starting point of omics analysis involves a crucial task of ensuring the presence of high-quality and clean data. However, accomplishing this task can be challenging due to the intricate nature of experimental platforms and workflows. There are several data quality-related matters that require attention. High-throughput technologies, such as microarrays, frequently encounter missing results and incomplete tests. In certain cases, it may be required to employ imputation techniques or omit affected samples. The presence of technical biases and batch effects might arise when experiments are conducted across different laboratories, times, platforms, and personnel. The application of normalization and regression techniques is necessary to mitigate these biases. The occurrence of noise and mistakes is commonly observed in mass spectrometry-based measurements of metabolomics and proteomics with low abundance. Therefore, implementation of filtering and quality control pipelines is recommended [26]. Efforts such as the Metabolomics Standards Initiative offer comprehensive recommendations for the establishment of reliable metadata and quality control measures in the field of metabolomics [44]. The ProteomeXchange collaboration enforces uniform standards among proteomics libraries that are part of its membership. It is imperative to proactively integrate comparable data quality frameworks throughout all omics domains [45].

4.2 Integration, interoperation, and federation are fundamental elements throughout diverse academic disciplines

The capacity for diverse systems, organizations, and disciplines to collaborate in a synchronized manner is of vital importance in the advancement of research and the acquisition of information. Nonetheless, the efficacy of such collaboration is frequently limited by differences in data formats, standards, and schemas [17]. In the domain of omics, the integration of data spanning genomes, transcriptomes, proteomes, and metabolomes is of utmost importance. However, this endeavor is not without its difficulties, mostly due to the heterogeneous nature of the data types involved and the diverse array of analysis platforms utilized. To address the problems mentioned earlier in the field of omics, various methodologies can be employed. Bioinformatics databases such as OmicsDI offer standardized APIs and data models [46], such as the Investigation-Study-Assay (ISA) framework, to streamline the retrieval of various omics datasets [47]. The PRIDE archive

and PeptideAtlas are examples of resources that collect proteomic data in standardized formats [48]. Controlled vocabularies, such as Gene Ontology (GO) and UniProt protein annotations, contribute to the establishment of a consistent semantic framework across many omics domains. The integration of these ontologies has been observed in numerous public repositories [49]. International consortia such as the Human Proteome Organization (HUPO) have developed federated databases [50], such as neXtProt, which serve as a knowledge platform for the integration of well-vetted proteomics data from many high-quality sources [51].

4.3 The subject matter under consideration belongs to the domains of ontologies and semantic definitions

The existence of varied nomenclature conventions, definitions, and metadata standards has a significant influence on the attainment of semantic interoperability. The application of ontologies and vocabularies in the formal representation of biological entities and their interactions facilitates a comprehensive understanding of information across diverse databases [30]. The Gene Ontology and Human Proteome Organization's PSI-OM are valuable resources that offer standardized annotation and descriptions for genes and proteins that have been found in various genomics, transcriptomics, and proteomics research [52]. Controlled vocabularies, such as the ontology developed by the Metabolomics Standards Initiative, serve to consolidate the nomenclature and categorization of metabolites based on their chemical structure and associated pathways [44]. Through the utilization of domain-specific ontologies, researchers in the field of omics can attain a common understanding and reduce uncertainty when integrating information derived from a variety of experiments. The utilization of ontologies also facilitates enhanced structuring of experimental metadata and the provision of formal descriptions pertaining to biological interactions. Therefore, to fully leverage the advantages of integrating omics data and achieve a more cohesive understanding of biological systems, it is imperative to establish and uniformly implement reporting standards and ontologies that prioritize semantic coherence [53].

4.4 Ensuring robust security measures, safeguarding privacy, and implementing ethical access rules

These are of utmost importance in handling sensitive biological data, particularly when obtained from human participants. The process of anonymization offers a certain degree of security; yet, it is important to acknowledge that inherent hazards still persist [54]. The dissemination of usage terms and conditions accompanied by informed agreement is critically important. In the context of aggregating and federating human omics data from many sources, it is imperative to establish a strong framework for data security and participant privacy to provide reliable governance. The implementation of encryption techniques and access control measures is crucial in safeguarding the confidentiality of participant data in the storage and transmission of sensitive genomics, proteomics, metabolomics, or other omics data. The process of anonymization or pseudonymization involves removing or replacing personal identifiers to minimize the danger of reidentification, hence preventing the tracing of data back to its original source. To ensure ethical practices, informed consent is obtained from individuals prior to the secondary use of their biosamples. Public database regulations are designed to facilitate restricted access to integrated datasets, ensuring that users are bound by data usage agreements that enforce ethical consumption. The provision of transparency and the ability to withdraw for participants empowers them with control over their personal information. Ensuring strict adherence to security best practices and ethics requirements is of utmost importance, particularly considering the widespread availability of human omics data resulting from prominent efforts such as the Precision Medicine Initiative [55], among others.

4.5 The incorporation of data analysis tools into databases

To facilitate continuous study of omics data, it is crucial to establish strong integration between databases, computational pipelines, and visualization tools. Cloud systems such as Terra, GenePattern, and SevenBridges offer seamless integration of genomics databases, such as TCGA, with popular analysis environments like Python and R [56]. There exist resources such as METLIN [57] and HMDB [58] that provide direct searching and processing of metabolomics data through web portals or locally employing Python and R packages. ProteINSIDE is a software tool that facilitates the interactive viewing and interpretation of proteomics data derived from many integrated sources [59], such as the Pride Archive [48] and PeptideAtlas [50]. The majority of omics repositories include public APIs that serve the purpose of enabling programmatic access to data and the development of customized analysis code and applications. Workflow platforms such as nf-core and Galaxy facilitate the integration of diverse databases and technologies, enabling the creation of customized analytic pipelines [60].

To address these difficulties, it is imperative to make progress in the field of database systems, as well as establish standards, ontologies, automation tools, and community agreements. It is very probable that the integration of advanced computing techniques, including parallelization, distributed processing, and machine learning, will bring about a significant transformation in biological databases. This transformation will enable such databases to evolve into intelligent discovery platforms that actively contribute to the advancement of systems biology research.

5. Future directions, including the role of machine learning and artificial intelligence in database querying and data integration

The future holds great potential for the utilization of AI in the domains of database querying and data integration. Machine learning (ML) algorithms have the capability to be utilized for the purpose of extracting valuable insights from databases and effectively integrating diverse data sources. ML holds promise in integrating diverse unstructured data sources. By acquiring knowledge about data formats, structures, and relationships, machine learning systems can consolidate interconnections among different sources that may appear unrelated. This approach will greatly simplify the integration of large amounts of data from multiple sources, like multiomics data. ML techniques, such as clustering, classification, and dimension reduction can help us understand complex and multi-dimensional data better. This allows us to ask questions and gather meaningful insights efficiently. ML can also automate the tasks involved in data preparation and cleaning. ML algorithms can address issues like missing values, duplicates, and inconsistencies automatically. This reduces the need for effort while improving the accuracy and reliability of the data. Furthermore, it has the potential to facilitate discovery within data warehouses and data lakes [61].

By incorporating natural language processing (NLP), we can enhance the user experience when interacting with databases by enabling natural query expressions. AI based systems utilizing NLP have the ability to transform text-based questions into query language (SQL) queries making these systems more accessible and user-friendly. This evolutionary process has the potential to expand database accessibility beyond professionals, reaching a range of individuals. By integrating AI capabilities into database management systems and query engines, we can enhance their data processing capabilities. ML-powered databases continuously improve their performance, ensure security, and expand their capacity in response to evolving data trends without intervention. However, it is still crucial to emphasize the importance of oversight to validate the results obtained from these systems [62].

The combination of automation and intelligence powered by ML and AI, along with NLP that enables more advanced user interfaces is predicted to have a significant impact on the future of database and data integration technologies. These technologies have the potential to accelerate data-driven exploration and support decision-making across domains. However, it will be crucial to manage the collaboration between humans and AI, to make this future a reality.

AI Disclosure

During the preparation of this work, the authors used ChatGPT. This tool has been used to correct spelling errors, enhance the flow of the sentences, and improve the language. After using this tool/service, the authors reviewed and edited the content as needed and takes full responsibility for the content of the publication.

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